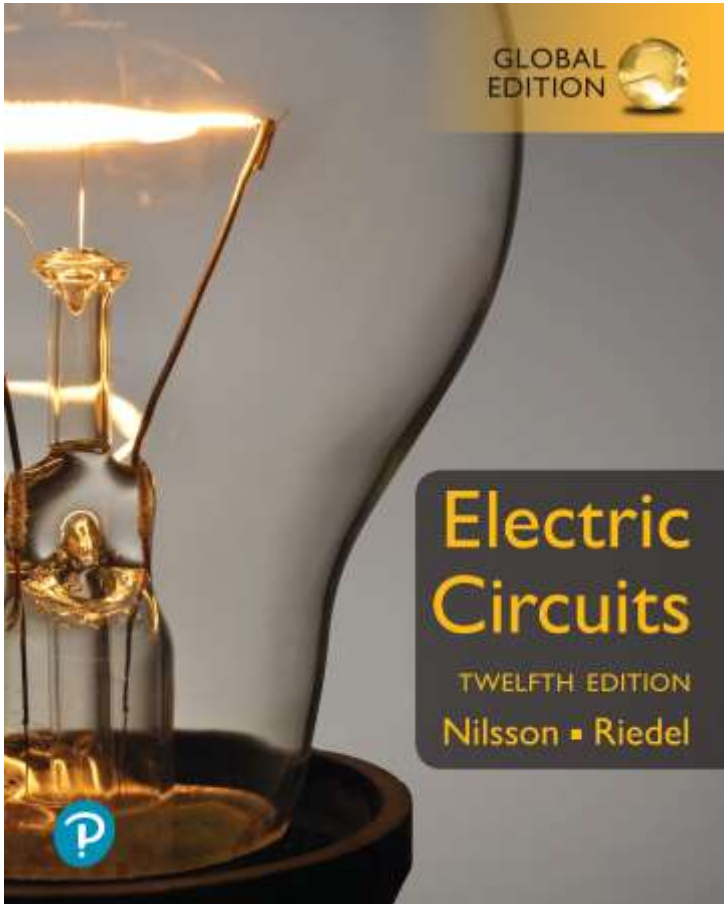
Electric Circuits

Twelfth Edition, Global Edition



Chapter 2 – Part 1

Circuit Elements

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Chapter 2 – Circuit Elements (1 of 15)

Objectives:

2.1 Understand independent and dependent sources

2.2 Understand resistors

2.3 Know and be able to use Ohm's Law

2.4 Know and be able to use the Kirchhoff Laws

2.5 Be able to calculate the power for a circuit

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Chapter 2 – Circuit Elements (2 of 15)

Electrical sources convert non-electric energy to electric energy.

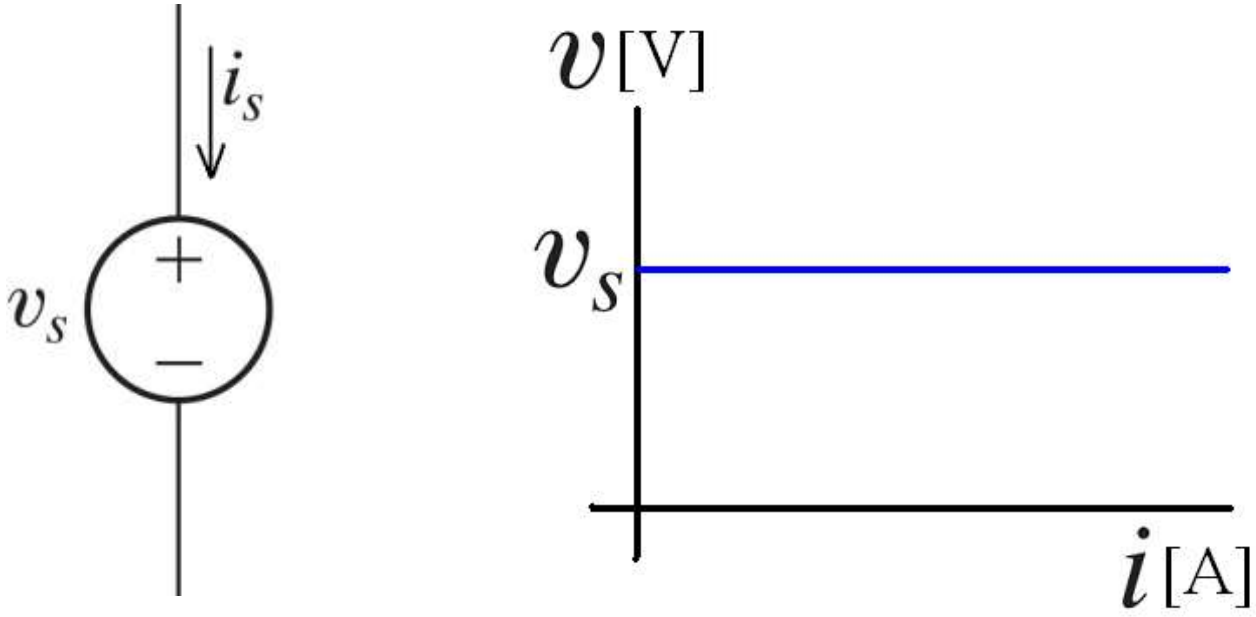
- Examples include
  - Batteries: convert chemical energy to electric energy
  - Generators: convert mechanical energy to electric energy
- We use 2 ideal models of electric sources:
  - Voltage sources, which supply the same voltage regardless of the current flow
  - Current sources, which supply the same current regardless of voltage drop

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Chapter 2 – Circuit Elements (3 of 15)

Ideal independent voltage source

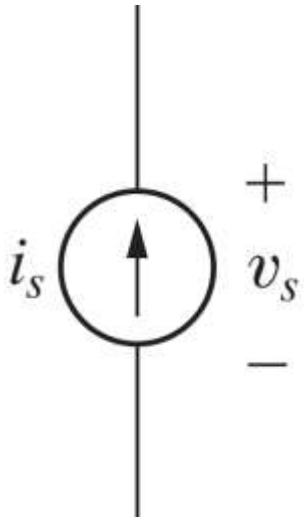


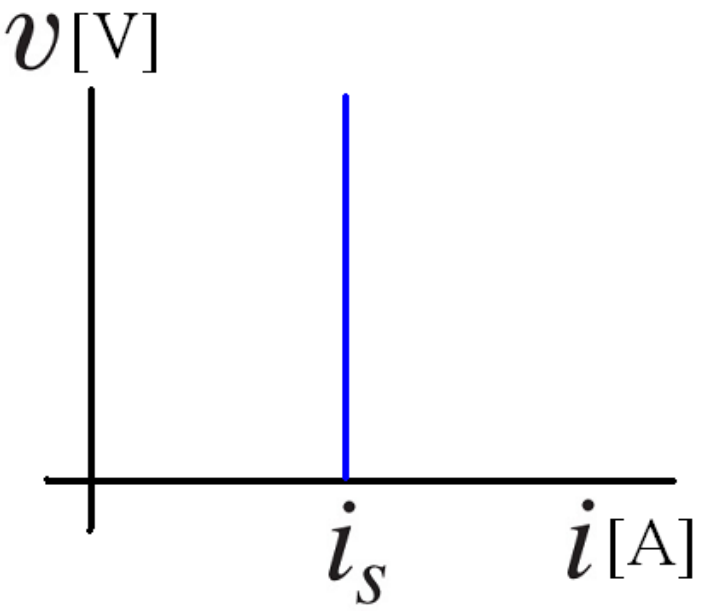
Observation – if all we have is the voltage drop across an ideal voltage source, there is no formula we can use to calculate the current through that source!

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Chapter 2 – Circuit Elements (4 of 15)

Ideal independent current source





Observation – if all we have is the current flowing through an ideal current source, there is no formula we can use to calculate the voltage drop across that source!

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Chapter 2 – Circuit Elements (5 of 15)

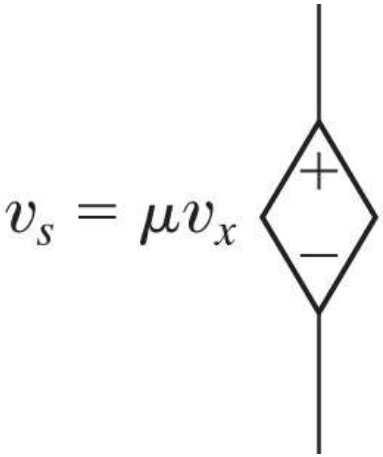
- The sources just presented are “independent” because the voltage or current supplied by the source does not depend on any other quantity in the circuit to which it is attached.
- As you might already have guessed, there are other sources that supply a voltage (or current) whose value depends on the voltage (or current) found elsewhere in the circuit.

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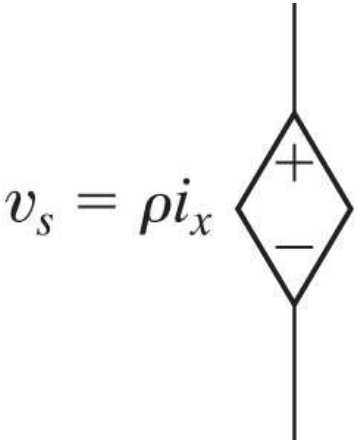
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Chapter 2 – Circuit Elements (6 of 15)

Dependent voltage sources



Voltage-controlled voltage source (VCVS), where v_s is the voltage supplied, v_x is the controlling voltage, and μ is the gain, whose units are V/V



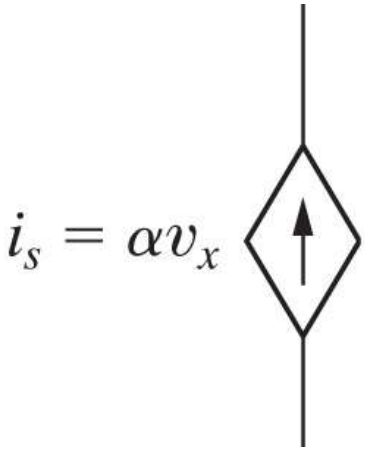
Current-controlled voltage source (CCVS), where v_s is the voltage supplied, i_x is the controlling voltage, and ρ is the gain, whose units are V/A

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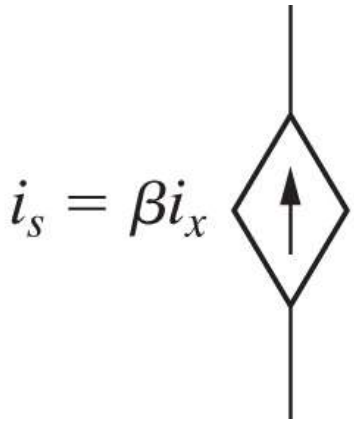
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Chapter 2 – Circuit Elements (7 of 15)

Dependent current sources



Voltage-controlled current source (VCCS), where i_s is the current supplied, v_x is the controlling voltage, and α is the gain, whose units are A/V



Current-controlled current source (CCCS), where i_s is the current supplied, i_x is the controlling current, and β is the gain, whose units are A/A

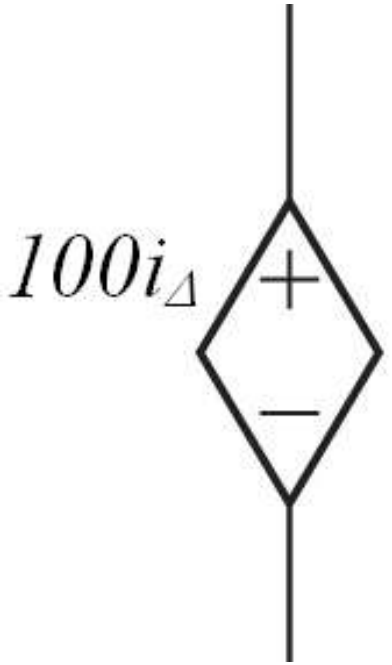
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Chapter 2 – Circuit Elements (8 of 15)

If i_Δ is 15 mA, how much voltage does this dependent source supply?

- ✗ A. 15 mV
- ✗ B. 1500 V
- ✓ C. 1.5 V
- ✗ D. 100 V



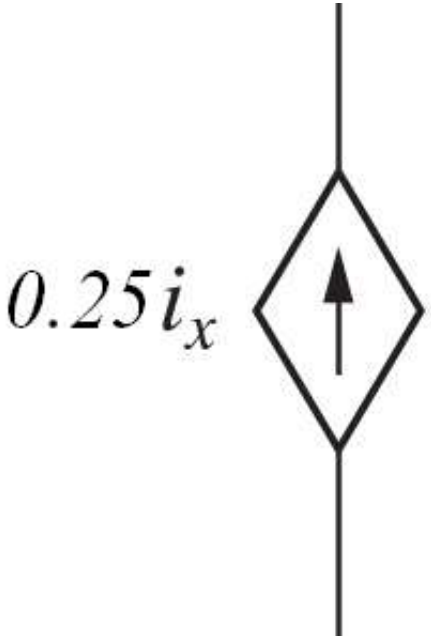
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Chapter 2 – Circuit Elements (9 of 15)

What units are associated with the gain 0.25 for this dependent source?

- ✗ A. V/V
- ✗ B. A/V
- ✗ C. V/A
- ✓ D. A/A



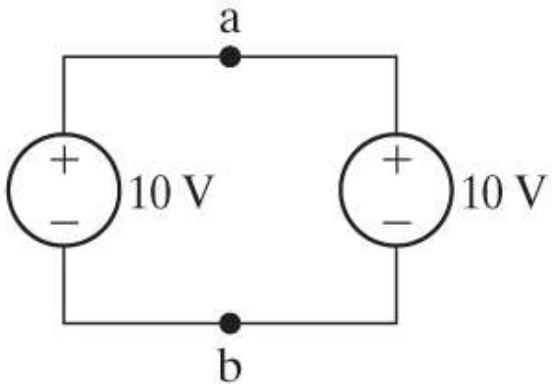
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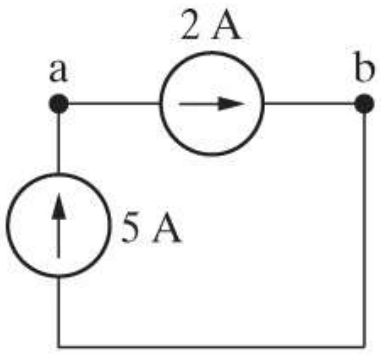
Chapter 2 – Circuit Elements (11 of 15)

Interconnected sources

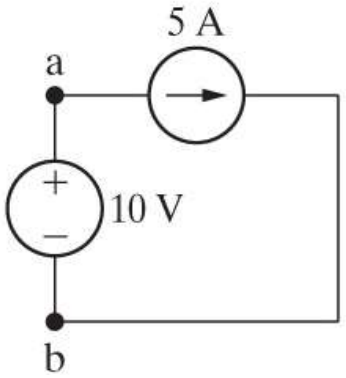
- When ideal sources are connected together to form a circuit, the connection is valid only if the constraints imposed by the ideal sources are not violated.



Valid or invalid?



Valid or invalid?



Valid or invalid?

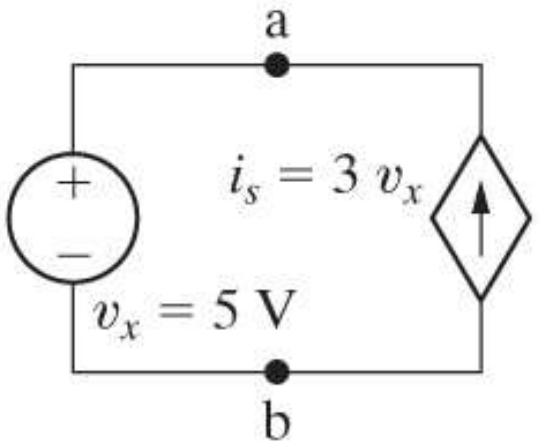
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Chapter 2 – Circuit Elements (12 of 15)

Is the interconnection shown here valid?

- ✔ A. Yes
- ✘ B. No

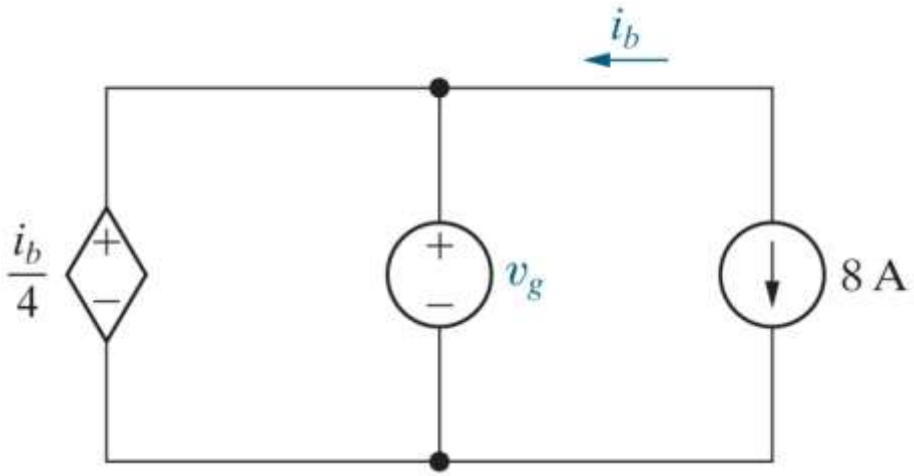


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Chapter 2 – Circuit Elements (14 of 15)

Example: Find v_g so that the interconnection is valid. Then find the power associated with the 8 A source.



$$v_g = i_b/4; \qquad i_b = -8\ \text{A}; \qquad \Rightarrow v_g = -8/4 = -2\ \text{V}$$

$$p_{8\text{A}} = v(8) = v_g(8) = (-2)(8) = -16\ \text{W}$$

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Chapter 2 – Circuit Elements (15 of 15)

We just discovered that the power associated with the 8 A source is −16 W. Is this source supplying or absorbing power?

- ✔ A. Supplying
- ✘ B. Absorbing

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